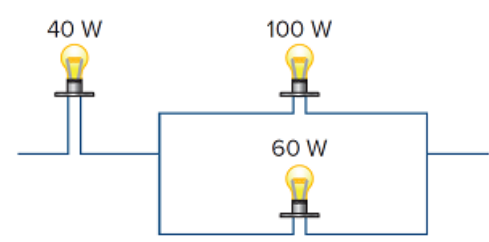


Problem

An enterprising young man travels to Europe carrying three light bulbs he had purchased in North America. The light bulbs he has are a 100-W light bulb, a 60-W light bulb, and a 40-W light bulb. Each light bulb is rated at 110 V. He wishes to connect these to a 220-V system that is found in Europe. For reasons we are not sure of, he connects the 40-W light bulb in series with a parallel combination of the 60-W light bulb and the 100-W light bulb as shown in Fig. 2.123. How much power is actually being delivered to each light bulb? What does he see when he first turns on the light bulbs? Is there a better way to connect these light bulbs in order to have them work more effectively?

Figure 2.123:



Step-by-step solution

Step 1 of 5

Refer to Figure 2.123 in the textbook.

Find the resistance of 40 W light bulb rated at 110 V.

$$\begin{aligned} R_{40\text{ W}} &= \frac{V^2}{P_{40\text{ W}}} \\ &= \frac{(110)^2}{40} \\ &= 302.5 \, \Omega \end{aligned}$$

Find the resistance of 100 W light bulb rated at 110 V.

$$\begin{aligned} R_{100\text{ W}} &= \frac{V^2}{P_{100\text{ W}}} \\ &= \frac{(110)^2}{100} \\ &= 121 \, \Omega \end{aligned}$$

Find the resistance of 60 W light bulb rated at 110 V.

$$\begin{aligned} R_{60\text{ W}} &= \frac{V^2}{P_{60\text{ W}}} \\ &= \frac{(110)^2}{60} \\ &= 201.667 \, \Omega \end{aligned}$$

Step 2 of 5

From the Figure, it is clear that $R_{100\text{ W}}$ and $R_{60\text{ W}}$ are connected in parallel. Find its equivalent resistance.

$$\begin{aligned}\frac{R_{100\text{ W}} \times R_{60\text{ W}}}{R_{100\text{ W}} + R_{60\text{ W}}} &= \frac{(121)(201.667)}{(121) + (201.667)} \\ &= 75.625 \, \Omega\end{aligned}$$

This resultant $75.625 \, \Omega$ is in series with $R_{40\text{ W}}$. Use voltage division rule to find the voltage across 40 W bulb.

$$\begin{aligned}V_{40\text{ W}} &= 220 \times \frac{302.5}{302.5 + 75.625} \\ &= 176 \, \text{V}\end{aligned}$$

Calculate the power delivered to 40 W bulb.

$$\begin{aligned}P_{40\text{ W}} &= \frac{(V_{40\text{ W}})^2}{R_{40\text{ W}}} \\ &= \frac{(176)^2}{302.5} \\ &= 102.4 \, \text{W}\end{aligned}$$

Therefore, the actual power being delivered to 40 W bulb is 102.4 W.

As the actual power is exceeds the rated power, the bulb gets burned.

Step 3 of 5

Find the voltage across combination of 100 W bulb and 60 W bulb as follows:

$$\begin{aligned}V_{40\text{ W}} + V_{\text{combination}} &= 220 \\ V_{\text{combination}} &= 220 - V_{40\text{ W}} \\ &= 220 - 176 \\ &= 44 \, \text{V}\end{aligned}$$

The voltage across 100 W and 60 W bulbs is equal. Therefore,

$$\begin{aligned}V_{60\text{ W}} &= V_{100\text{ W}} \\ &= 44 \, \text{V}\end{aligned}$$

Calculate the power delivered to 60 W bulb.

$$\begin{aligned}P_{60\text{ W}} &= \frac{(V_{60\text{ W}})^2}{R_{60\text{ W}}} \\ &= \frac{(44)^2}{201.667} \\ &= 9.6 \, \text{W}\end{aligned}$$

Therefore, the actual power being delivered to 60 W bulb is 9.6 W.

Step 4 of 5

Calculate the power delivered to 100 W bulb.

$$\begin{aligned}P_{100\text{ W}} &= \frac{(V_{100\text{ W}})^2}{R_{100\text{ W}}} \\ &= \frac{(44)^2}{121} \\ &= 16 \, \text{W}\end{aligned}$$

Therefore, the actual power being delivered to 100 W bulb is 16 W.

Step 5 of 5

On connecting 40 W bulb in series with the parallel combination of 100W and 60 W bulb to a 220 V supply, when switch on the circuit, the 40 W bulb cannot withstand as the power delivered to it is more than the rated power and hence it blows out.

The best way to connect this circuit is 100 W bulb is to be connected in series with the parallel combination of 60 W bulb and 40 W bulb. Then there will not be any damage to the bulbs and the circuit will function properly as desired.